Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2022-2023

Comprehensive Examination
(EC-3 Regular)

Course No.	: DSECLZG522
Course Title	: Big Data Systems
Nature of Exam	: Open Book
Weightage	: 40%
Duration	: 2.5 Hours
Date of Exam	: 14-October-2023 (FN)

No. of Pages = 2
No. of Questions = 8

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made, if any, should be stated clearly at the beginning of your answer.

Q1. Why is it recommended to use HDFS for storing large volume data files but not when there are lots of small files? Describe the scenarios in which you set different block sizes for files in HDFS?

(4 Marks)

Q2. What limitation of Hadoop is getting addressed in HBase? HBase persists data in HTables in HDFS and Cassandra stores data in SSTables. HTables and SSTables are immutable. Explain how updates are deletion of rows is implemented in HBase and Cassandra?

(5 Marks)

Q3. You are a data engineer working for a large e-commerce company that uses Hadoop and its ecosystem for data processing and analysis. Your company is dealing with a massive amount of transaction data stored in HBase tables, and you need to extract and analyze this data using Hive and Pig. Additionally, you want to move data between Hadoop and a relational database using Sqoop. Your goal is to optimize data processing tasks efficiently.
You have an HBase table containing transaction data with the following structure:
- Column Family: "info"
- Columns: "user_id," "product_id," "timestamp," "amount"
You need to write a Pig script to find the total sum of "amount" for all transactions. Assume the HBase table is named "transactions."

(6 Marks)

Q4. You are designing a NoSQL database for a social media platform. Each user can have multiple followers, and each follower can follow multiple users. How would you model this relationship in a NoSQL database, and which type of NoSQL database type (e.g., document-oriented, Key-Value, Graph, Column-family) would you choose for this scenario? Provide a schema and give justification for your choice.

(5 Marks)

Q5. Go through the Spark Scala code given below and answer the questions (a,b,c,d) given below:

```
1   val lines = new Array[String](2)
2   lines(0) = "one one reject"
3   lines(1) = "own own own"
4   val stringRDD=sc.parallelize(lines,2)
5   println("Number of Partitions: "+stringRDD.getNumPartitions)
6   val wordsRDD = stringRDD.flatMap(x => x.split(" "))
7   val wc = wordsRDD.map(word => (word, 1)).reduceByKey(_ + _)
8.  wc.collect()
9.  val filt1RDD = wordsRDD.filter(x => x.contains("own"))
10. filt1RDD.collect()
```

Questions:

a. What is the output at line No. 5

b. What is the output at line No. 8

c. What is the output at line No.10

d. Modify the code to find out the word count after excluding the word "reject"

(4 Marks)

Q6**.** You have a large dataset with customer reviews, and you want to use Spark to perform the following tasks:

- Count the total number of reviews.
- Find the average rating.
- Determine the number of reviews for each product category.

Read the “reviews.txt” file into an RDD and write a Spark code snippet (in Python or Scala) to accomplish these tasks.

(5 Marks)

Q7. Why is windowing needed in Stream computing? Data from a smart device is transmitted over TCP/IP connection to a Spark streaming analytics system. Data from the device arrives at the rate of 1 packet per second. The batching interval is set to 5 seconds. The stream analytics system uses a stateful algorithm with a Time window of 60 seconds. It is required to generate an output from the streaming system every 30 seconds. What type of windowing is recommended by you? Give appropriate configuration parameters for the Window suggested by you. What will be the number of RDDs in the Dstreams generated in one window duration.

(5 Marks)

Q8. Explain how the features of cloud native NoSQL databases help in the design and deployment of Bigdata applications on Cloud?

(6 Marks)

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